



Unit 4 · Outcome 3 · Scientific Investigation

Findings & implications

KK12

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



The "so what?" moment

An investigation that stops at a graph is only half-finished. The marks live in two sentences: what you found, and why it matters.

Imagine you spend a weekend baking, taste-testing and recording which oven shelf browns a cake most evenly. If a friend asks "how did it go?" and you just hand them a spreadsheet of temperatures, they'll stare at you. What they want is the finding ("the middle shelf browns most evenly") and the implication ("so put cakes on the middle shelf — and it probably applies to biscuits too"). Science is the same. The data is evidence; the finding and implication are the payoff.

In a VCE Environmental Science scientific poster, this is the job of the Discussion and Conclusion. The Discussion interprets the data and links it back to the question; the Conclusion answers the aim in one or two crisp sentences. Implications then push outward: what does this mean for management, for other locations, for the wider field?

↑ The closer to the centre, the more certain you are. The further out a ripple travels, the more carefully it must be justified by the evidence behind it.

Result → finding → conclusion → implication

These four words are not synonyms, and examiners deliberately test whether you can tell them apart. Think of them as a relay — each one passes the baton a step further from the raw numbers.

Term	What it is	Priya's creek, worked
Result	A raw, specific number or observation straight from your data — no interpretation.	"Mean turbidity at 0 m was 41 NTU; at 175 m it was 12 NTU."
Finding	The pattern or relationship the results show, in plain interpreted language.	"Turbidity decreased steadily with distance downstream of the outfall."
Conclusion	The answer to the aim/hypothesis, stated with the level of support the evidence gives.	"The data supports the hypothesis: the stormwater outfall raises turbidity in Wirrim Creek nearest the outfall."
Implication	The "so what?" — meaning beyond this study: for management, other sites, theory, or future work.	"A sediment trap at the outfall could reduce downstream turbidity; similar outfalls on urban creeks may need monitoring."

✓ TIP

The one-line test A result has a number. A finding has a trend word (increased / decreased / no difference). A conclusion mentions the hypothesis or aim. An implication mentions something outside the study (management, other locations, future research).

Writing a key finding

A key finding is the headline relationship your investigation uncovered. A strong one does three things at once:

- Names the relationship between your independent and dependent variables (what changed with what).
- Uses the direction word — increased, decreased, was higher/lower, showed no difference.
- Anchors it in quantified evidence — a value, a range, or a comparison from your own data.

◆ KEY INSIGHT

Finding formula As [independent variable] [changed], [dependent variable] [direction] (from X to Y), indicating [relationship]. e.g. "As distance downstream increased from 0 to 175 m, mean turbidity decreased from 41 to 12 NTU, indicating the outfall is the source of suspended sediment."

Notice what a finding is not: it is not a re-description of your method ("I measured turbidity at eight sites"), and it is not a raw data dump ("the values were 41, 38, 30..."). It is the interpreted pattern — the sentence a reader could not have written without your graph in front of them.

The implication radius

Implications spread outward from a finding like ripples. The trick is to recognise the different directions meaning can travel — examiners reward breadth here. Explore the five domains below.

Direction	Question it answers	Priya's creek
Environmental	What does this mean for the ecosystem itself?	High turbidity reduces light, harming aquatic plants and sediment-sensitive macroinvertebrates near the outfall.
Management	What action could follow?	Install a sediment trap or revegetate the bank to filter stormwater before it reaches the creek.
Generalisation	Where else might this apply?	Other urban creeks with stormwater outfalls may show the same gradient — worth monitoring.
Scientific / theory	How does it fit known science?	Supports the model that urban runoff is a point source of suspended sediment, consistent with catchment hydrology.
Further research	What should be investigated next?	Sample across seasons or during rain events to test whether the effect intensifies after storms.

• NOTE

Why breadth scores An extended-response question asking for "implications" rarely wants one. Listing two or three different directions (e.g. an ecological consequence and a management response and a next step) shows you understand that findings matter in more than one way.

Staying inside your evidence

The most common way to lose implication marks is to over-reach — to claim more than your data can support. A finding from one creek, on one day cannot prove a fact about every waterway on Earth. The skill is matching the scope of your claim to the scope of your evidence.

- "This proves ..." — a single student investigation supports or is consistent with; it rarely proves.
- "...in all rivers / every creek" — you sampled one site; generalise cautiously ("may also occur in similar urban creeks").
- "...is definitely caused by ..." — correlation found; unless you controlled other variables, say "is consistent with" or "suggests".

✓ TIP

Hedge like a scientist Words such as suggests, indicates, is consistent with, may, could, within this study are not weakness — they are precision. They tell the examiner you know exactly how far your evidence reaches.

Command words

◆ COMMAND WORDS

The verb in the question controls how much "so what?" you owe. Tap each to reveal what it demands.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

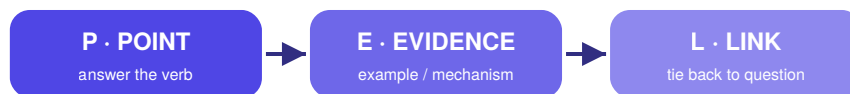
Match your answer to the verb — the fastest marks in the exam.

Exam traps

⚠ EXAM TRAPS

The recurring ways students leak marks on findings-and-implications questions:

Writing trainer · P·E·L



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

🔗 P·E·L WRITING

Strong implication answers follow P·E·L — P oint (state the implication), E vidence (cite your own data), L ink (connect to management, theory or the wider world). Draft a response below; the meters light up as you hit each move.

Weak vs strong

Same prompt — "State one key finding and one implication of Priya's investigation (4 marks)" — answered two ways. Read the annotations.

Practice exam

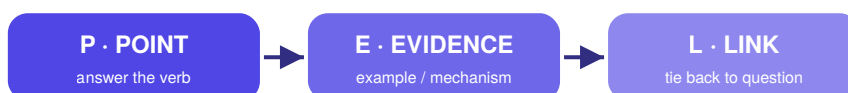
★ PRACTICE & SELF-MARK

Five graded questions, 20 marks total. Attempt each, reveal the marking scheme, then self-mark — your score collects in the dock.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Priya found that the turbidity was high. This proves that all stormwater drains pollute every river, so the council should ban all drains." No number or relationship — "high" is not a finding, it's vague. "Proves... all rivers" massively over-reaches one creek on one day. The implication ("ban all drains") is not justified by the evidence and isn't realistic management.

✓ STRONG / MODEL ANSWER

A key finding was that mean turbidity decreased from 41 NTU at the outfall to 12 NTU at 175 m downstream, indicating the outfall is the main source of suspended sediment. Finding = relationship + direction + quantified evidence. ✓ One implication is that installing a sediment trap or revegetating the bank could reduce turbidity and protect light-sensitive aquatic life near the outfall. Implication links the finding to a realistic management action and an ecological consequence. ✓ As only one creek on one day was sampled, this suggests rather than proves the effect, and similar urban creeks would warrant monitoring." Hedged scope + a generalisation/further-research impli